

New Approaches to Groundplane Measurements

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Loudspeaker designers rely on acoustical measurements to evaluate individual drivers and overall system performance. Among these techniques, groundplane measurements are widely popular with amateurs and professionals for capturing free-field data. However, because the floor sits in such close proximity to the cabinet, the groundplane's influence on a loudspeaker's in-room response cannot be ignored. This paper introduces a new hybrid measurement method that offers several advantages over existing techniques and provides more complete characterization of the loudspeaker. Additionally, it presents new correction methods for groundplane measurements conducted on surfaces with imperfect acoustic reflectivity.

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Background

Consider figure 1, depicting a loudspeaker set up for normal listening conditions on a groundplane. Sources within the loudspeaker produce sound that travels towards an observer located on the listening axis, at approximately the tweeter height. The “direct” or “through air” output follows the shortest path between source to observer (e.g. figure 1, dotted line). Output also leaves each source at some off-axis angle that allows it to travel first towards a reflection point on the groundplane and then onwards to the observer. There is only one possible through-air path for the direct sound – a straight line from source to observer. Likewise, there is a unique path via the groundplane to the observer from each source because the reflection from the groundplane is specular (incoming and outgoing angles are equal). The direct sound will arrive before the reflected sound because the reflected sound travels a longer path. As a result, the magnitude of the reflected sound will be lower and, due to the delay, will have additional phase rotation.

The loudspeaker designer would like to know what the observer “hears”, and so would like to be able to perform a measurement at this location, however, when any portion of the time record that contains both direct and reflected signals is converted into a frequency response there will be an interference pattern produced. An example of such a pattern is given in figure 2. This is the predicted response for a small, baffled source with directivity placed 1m above a reflective groundplane similar to the tweeter position in figure 1. In the absence of the reflection e.g. in free-field conditions, the frequency response of the source would equal 0dB for all frequencies. When the reflection is also present, it adds to the direct sound coherently at low frequencies, raising the level by

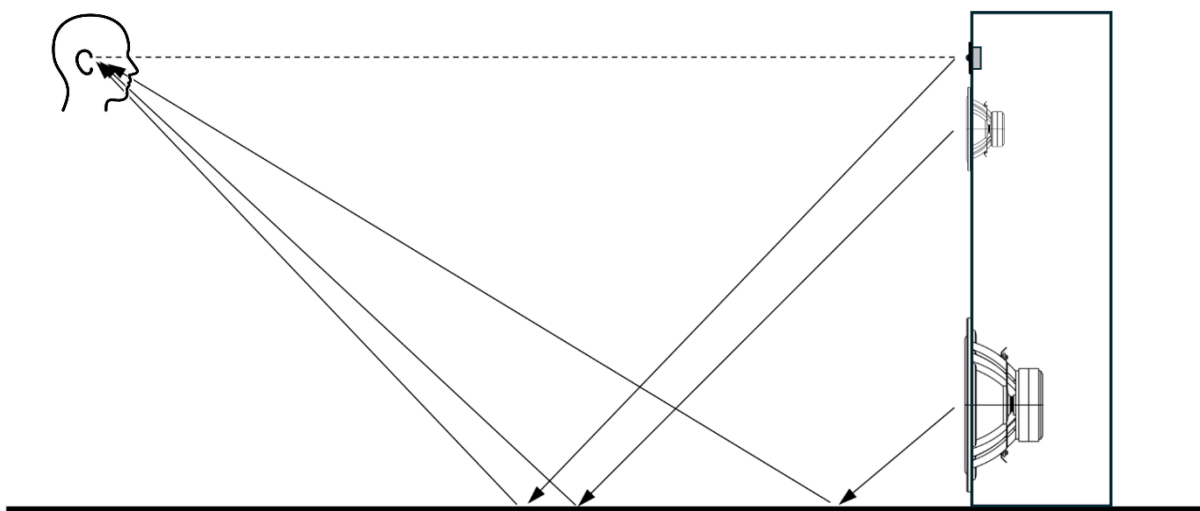


Figure 1 – A loudspeaker resting on a groundplane. Both direct and reflected output from the loudspeaker reaches the listener.

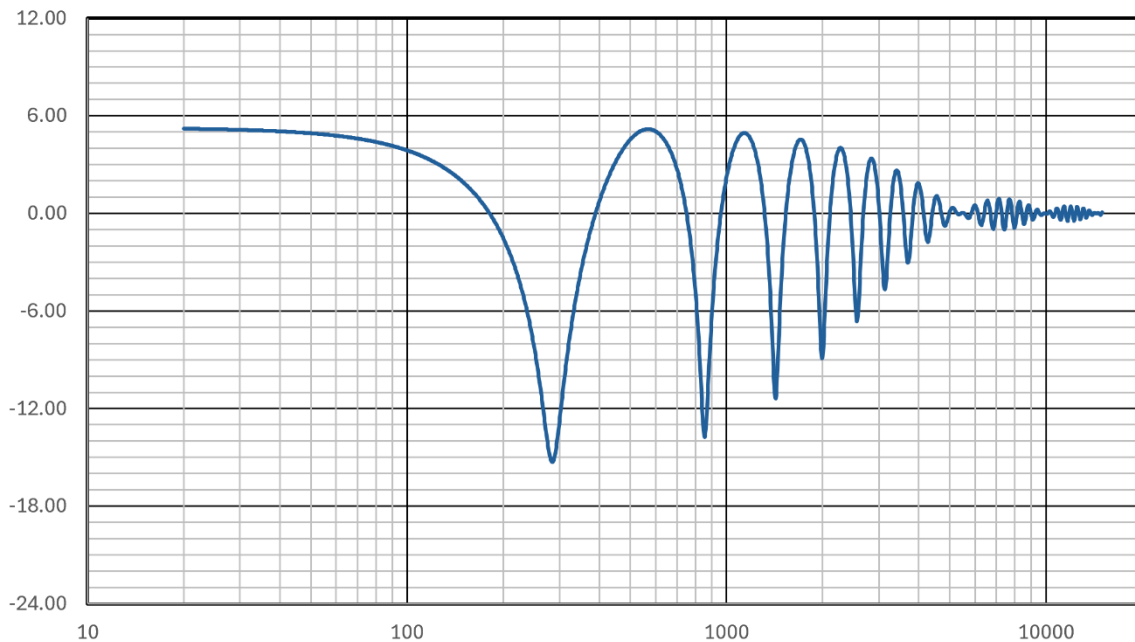


Figure 2 – The frequency response of a baffled, directive source showing the interference pattern that is produced when the groundplane reflection interacts with the direct sound at the listening position.

almost 6dB. At middle frequencies it causes interference as the phase difference between direct and reflected output begins to be significant. At high frequencies the increasing directivity of the source weakens the magnitude of the off-axis output, and the level drops back down to 0dB, consisting primarily of the direct sound. When the goal is to obtain the response of only the source, the interference pattern represents too much uncertainty in the measurement. One would like to use measurement conditions that will eliminate or not produce any interference so that the source can be properly characterized independent of the effect of the immediate environment. At the same time and because of the different ways that the reflection interacts with the direct sound, understanding how the in-room response differs from the free-field response represents valuable information to the designer.

It is well known that it is counter-productive to make in-room measurements at the listening position. If one instead places a microphone on the groundplane will make the direct and reflected pathlengths nearly identical and will eliminate nearly all of the frequency domain interference. Unfortunately, because the pickup end of the microphone has a finite size, one cannot place it exactly on the groundplane. Gander [1] shows that this limits the high frequency measurement accuracy due to the small pathlength difference that remains, and the higher the source or microphone elevations the worse this effect will be. Gander suggests tipping the loudspeaker towards the microphone and placing the tweeter as close to the groundplane as possible to

minimize this effect. Another option is to place the loudspeaker on the ground on its side. When the reflection no longer causes interference, the resulting response is much like what would be gathered under free-field conditions.

Loudspeaker designers often use the free-field response to design the crossover network, however, a loudspeaker's in-room and free-field responses can be quite different and it can be instructive to learn how the groundplane reflection will impact the frequency and power responses when the loudspeaker is set up for listening. Rather than designing the loudspeaker and crossover using free-field data, some loudspeaker systems incorporate boundary reflections directly into the design [2]. It is often assumed that the groundplane will simply add 6dB of gain to the loudspeaker, however, the exact effect in the frequency domain is much more complicated, as shown in figure 2. Understanding how the groundplane reflection influences the in-room response and the perception of tonal balance by the listener will better position the designer for success. Because a deeper discussion requires a review of hearing and psychoacoustics, I will instead provide the reader with a few references [3,4,5] with which they can begin to explore that topic. Instead, we will focus on acoustic measurements that both make use of, and at the same time eliminate, the groundplane reflection of the loudspeaker.

Theory

Consider again figure 1, where an observer (or microphone) is located above the groundplane. An equation that describes the sound pressure is:

$$p_{TOT} = p_D + p_R = \frac{e^{ikr_1}}{r_1} + R \frac{e^{ik(r_2+r_3)}}{(r_2 + r_3)} \quad (1)$$

This is equation 8 from Campos and Olivera [6] with slight changes to the nomenclature. Although the authors developed the formalism for a simple source, the general form of the equation applies equally well to any source type. The total pressure (p_{TOT}) is expressed as a sum its of direct (p_D) and groundplane reflection (p_R) components. In the equation, r_1 is the direct, through-air distance between source and microphone, r_2 is the distance between source and the reflection point on the groundplane, and r_3 is the distance onward from the reflection point to the microphone.

In the expansion of the p_R term, "R" represents the transfer function of the groundplane reflection. The transfer function represents the effect that the reflection process has on the wavefront. R can have has a magnitude between 0 and 1 and may also have a phase component. The lower the value the weaker will be the outgoing reflected acoustic wave compared to the incoming wave. For a perfectly hard, flat, and smooth surface this

would have a value of $1.0 + 0i$. An example of such a surface is a very smooth poured concrete floor. Materials such as hardwood like in a basketball gymnasium floor have a value of around 0.9 to 0.95. When the groundplane is a relatively soft material the value can decrease below 0.5 and will be frequency dependent.

In addition to the mathematics of the system, understanding certain properties of the groundplane reflection will be useful later. Recall that the groundplane reflection is “specular”. This term means that it acts like a mirror such that the groundplane does not scatter an incoming wave in many directions. For specular reflectivity, the incoming and outgoing angle of the reflection are identical, and we can re-arrange the geometry and view the system from a different perspective. In figure 3 a second “upside-down loudspeaker” appears in the cartoon. The second loudspeaker is the “image” of the actual real-world loudspeaker that is sitting on the groundplane. This is a different but equally valid way to represent the arrangement of figure 1 - the loudspeaker and its image are suspended in free space and the microphone receives direct sound both from the source (D) as well as the image (I). This arrangement of “source plus image” that is shown in figure 3 will produce the same response as a real-world loudspeaker sitting on a reflecting groundplane that is depicted in figure 1. In figure 3, instead of output from the real loudspeaker reflecting off of the groundplane, direct output from the image (I) passes through a plane (at R) where the groundplane was located in

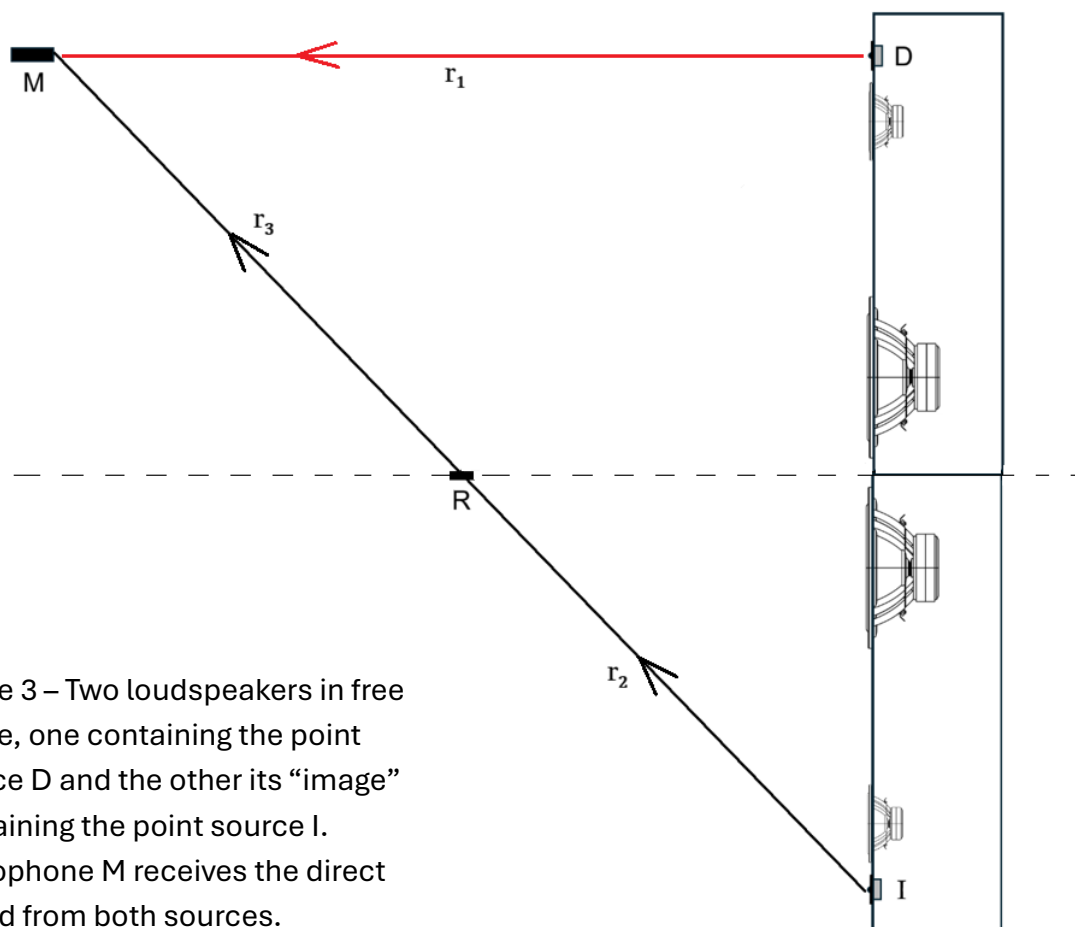


Figure 3 – Two loudspeakers in free space, one containing the point source D and the other its “image” containing the point source I. Microphone M receives the direct sound from both sources.

figure 1 en route to the microphone (M). There is no “ground” per se, but rather just an imaginary boundary plane. By re-framing the geometry in this way, we can describe the task of removing the reflection as removing the output or signal component that is produced by the loudspeaker’s groundplane image.

Figure 3 illustrates a second effect of the groundplane on acoustic measurements of the loudspeaker. Because the real-world loudspeaker and its image in the groundplane are touching, the acoustic support provided by the baffle is now twice as large and the baffle-step response will be different than the single loudspeaker in free-space. This is sometimes called “baffle doubling”. Because the loudspeaker is always in proximity to the groundplane (the floor) under normal listening conditions, this effect is always present. On the other hand, measurements that result in a free-field or quasi-free-field response completely lack this effect because the contribution from the reflection has been completely eliminated. Similarly, measurements performed using the method of Gander will also be in error when the loudspeaker is placed on its side or upside down because these orientations with respect to the ground plane are not present under normal use conditions. Although it may seem possible to manipulate the effect of the groundplane image by modeling the various responses of single and doubled loudspeaker baffle, diffraction modeling software is not always reliable and accurate to better than 2dB. Better would be a technique that can produce both the free-field and in-room responses from a typical “in-room” setup configuration.

So far in this section we have only described the output from a single point source on the loudspeaker. The complete output of the loudspeaker can be represented by an

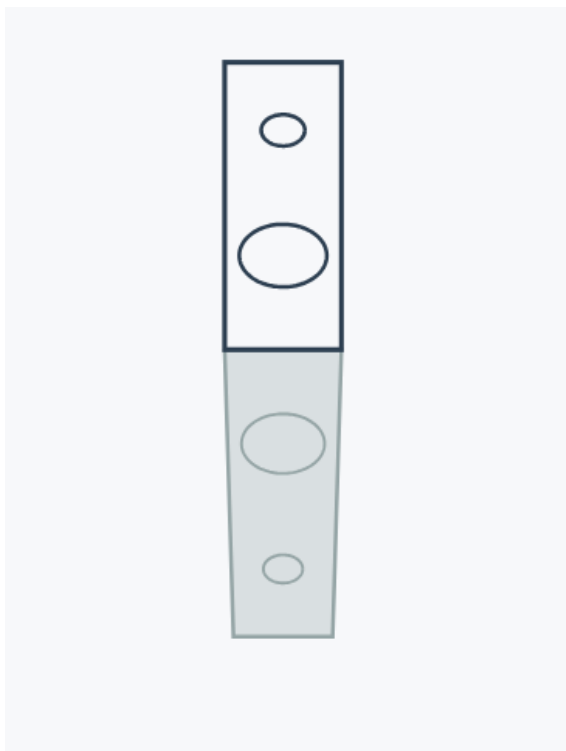


Figure 4 – The real-world loudspeaker and the region occupied on the groundplane by its image, when the loudspeaker is resting on the floor, as seen by the microphone.

ensemble of point sources – the points that make up the radiating surface of each driver, secondary sources along the baffle edge, etc. Due to the constraints of specular reflection, each point source will have a slightly different reflection point on the ground, and the ensemble of these points on the ground is the “image” seen from the vantage point of the microphone in the groundplane (see figure 4). As you move the position or elevation of the microphone or the loudspeaker the apparent size of both the loudspeaker and its image will change. The relative distance and thus the phase rotation of any point source making up the loudspeaker will also change, as seen from the observation point. A more familiar way to state this is that the response measurement is position dependent, however, it is important to understand the concept of the groundplane image as seen from the microphone, and how points on the loudspeaker map to points within the image.

Groundplane Measurement Techniques

A measurement technique involving a reflecting boundary was described as early as 1930 by Malter and also by Kellogg, and more recently by Mark Gander in his 1982 JAES paper. A useful modification to the Gander technique will now be described and then a completely new hybrid measurement technique will be introduced.

Lift and Tuck

The first technique I will call the “lift and tuck” groundplane technique. This is similar to, and an improvement on, the method of Gander. Instead of placing the loudspeaker on the groundplane, it is elevated (the lift) and then rotated or tilted downwards (the tuck) to point at the microphone, which sits on the groundplane (see figure 5). Because the loudspeaker is elevated above the groundplane, as seen from the microphone there is an apparent separation between it and its image that is equal to twice the height of the stand. As a result, the loudspeaker and its image are not acoustically close enough for their outputs to couple except at rather low frequencies. It has been commented that if you elevate the loudspeaker about half the distance that a pair of loudspeakers would be laterally separated in a room, the remaining coupling in the low bass represents what will occur laterally between two speakers under normal in-room listening conditions.

Often the loudspeaker is placed horizontally on the stand instead of vertically, since this is a more practical and stable orientation for the mass of the loudspeaker. The groundplane position of the microphone should be aligned with the listening axis of the loudspeaker. By varying the tuck angle while the microphone remains stationary, off axis measurements (in the loudspeaker’s horizontal plane) can be obtained. The physical support used to elevate the loudspeaker should be acoustically transparent. When the



Figure 5 – An example of a loudspeaker on a stand that the practitioner has set up for the lift and tuck technique. The microphone is out of frame, to the right. Note that the groundplane surface consists of smooth poured concrete.

groundplane is hard and smooth and large in lateral extent and the microphone is small and in contact with the groundplane the Lift and Tuck method can produce very accurate results up to approximately 6k -10k Hz where it is limited by the effect described by Gander in the appendix of his paper.

With the microphone placed on the groundplane, the output from loudspeaker and its image in the groundplane are recorded by the microphone like in figure 3, but now the microphone is located equidistant from them. This means that, for all but the highest audio frequencies, the phase angle will be identical and the outputs will add coherently, increasing the SPL level by 6.02 dB (see note 1).

The data produced by the lift and tuck technique does not include the groundplane baffle doubling that will occur under normal use, however, it is straightforward to perform and when measurements are taken on a reflective groundplane the results are easily converted to the free-field response of the D.U.T. This response is typically used to design and confirm the crossover network for a loudspeaker and check the driver integration. As such it provides useful information to the designer and involves test conditions that are relatively easy to achieve.

Measure, Move-Move, Measure

By considering how to isolate the reflection component produced by the groundplane as seen by the microphone, a new technique for removing it can be synthesized. First, consider the arrangement shown in figure 6. It would be possible to place the microphone on the groundplane at R, the reflection point, and make a groundplane measurement. At R the microphone will capture the off-axis output that continues past it and up to the listening axis microphone position. The data could be manipulated to mimic the propagation along the distance r_3 to the microphone, and this would seem to be a viable way to obtain the reflection part of the signal. Unfortunately, that will only be accurate for a single point source – the one with a reflection at the point where the microphone is located. Recall that the loudspeaker's complete output can be thought of as an ensemble of point sources and, since the sources are all at different positions and especially different heights above the ground, the ensemble of reflection points occupies an area on the ground, not a point. Placing the microphone on the ground at one position will not sample the groundplane image in the same way as the microphone placed at the listening position. Also, because the microphone at R is a different distance away from the loudspeaker when the microphone is at M, the perspective on it is different and this would produce relative errors in the phase components for the point sources making up the loudspeaker. These sorts of errors are eliminated in the Measure, Move-Move, Measure Technique.

The Measure, Move-Move, Measure technique combines direct and groundplane measurements. It can produce not only the free-field response of the loudspeaker, but also the groundplane reflection response and the in-room response showing the frequency dependent gain that the groundplane reflection generates. The technique

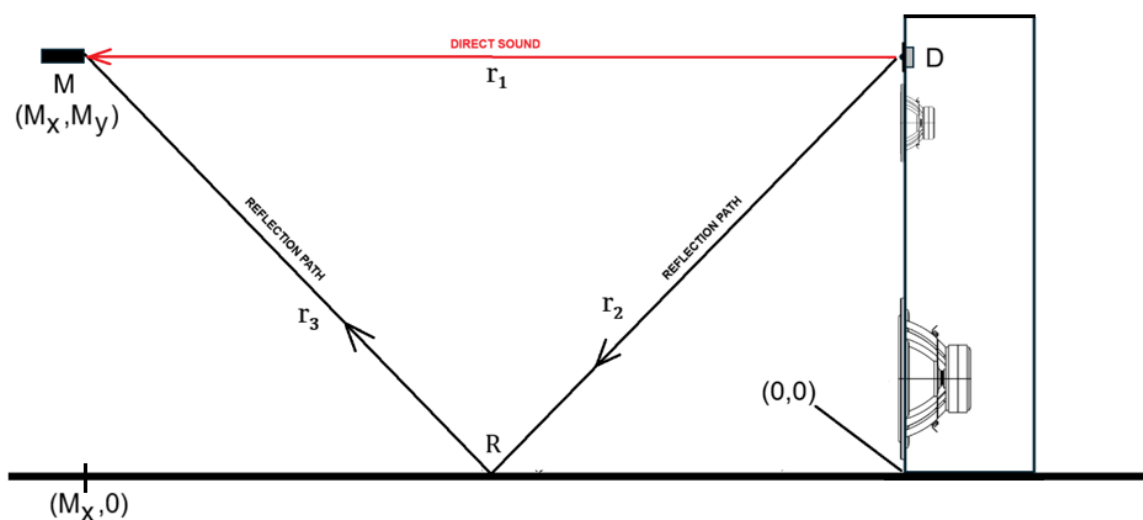


Figure 6 – The setup for measurement #1 of the Measure, Move-Move, Measure technique. The DUT rests on the ground and the microphone is on the listening axis.

requires two separate measurements, each with a different position of microphone and loudspeaker:

MEASUREMENT #1: The first measurement is made exactly as shown in figure 6. The microphone (M) is placed at position (M_x, M_y) on the listening axis and at some standoff distance and the loudspeaker is placed upright on the ground and aligned with the coordinate system origin $(0,0)$. The microphone records the direct sound and the groundplane reflection from all the point sources that make up the loudspeaker output. The red and black lines in figure 6 illustrate these for one such source.

To arrive at the conditions for measurement #2 we will need to perform some position gymnastics. Start with figure 6 and reflect the microphone position about the ground plane, e.g. from (M_x, M_y) to $(M_x, -M_y)$. This new arrangement is illustrated in figure 7. Compared to the reflection component shown in figure 6, the microphone is positioned at the same off-axis angle and at the exact same distance from the source, $r_2 + r_3$. This would be a nice place to make a second measurement except that the microphone

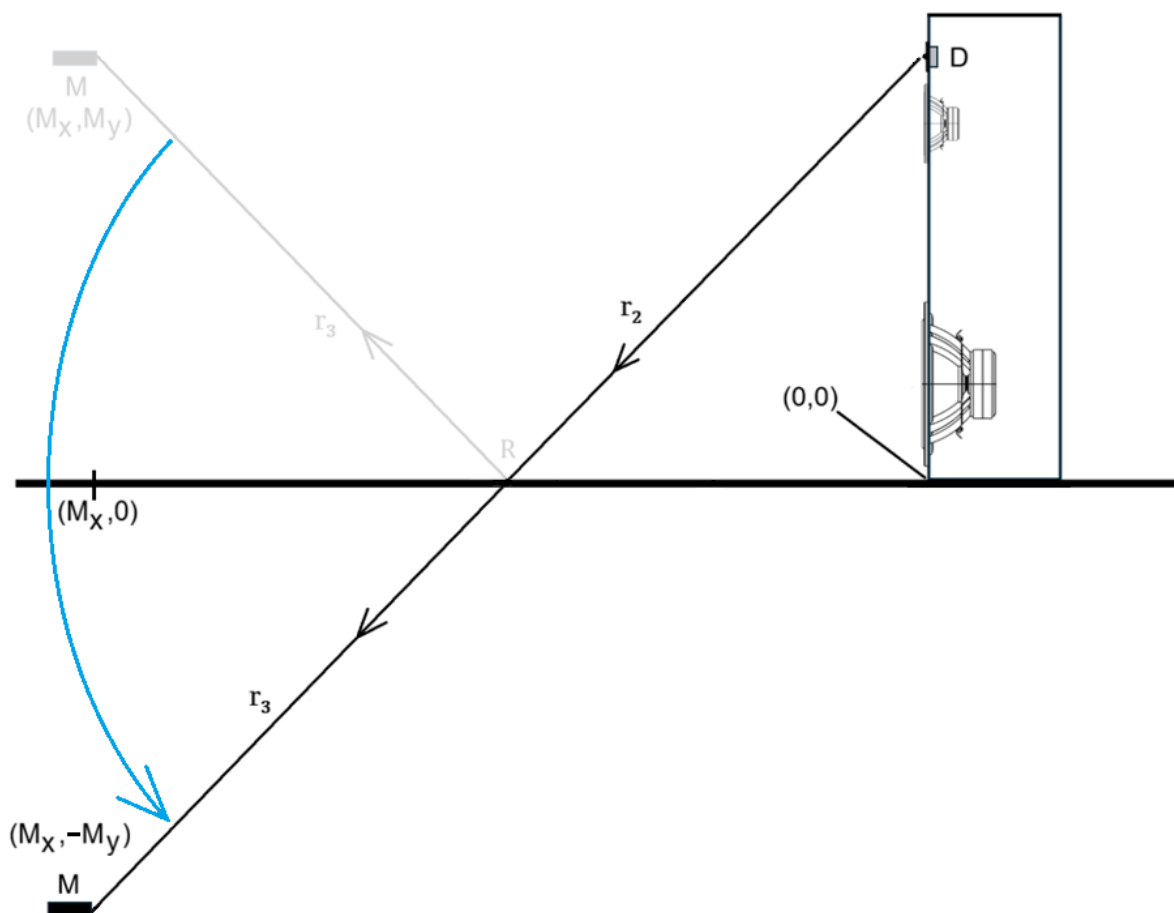
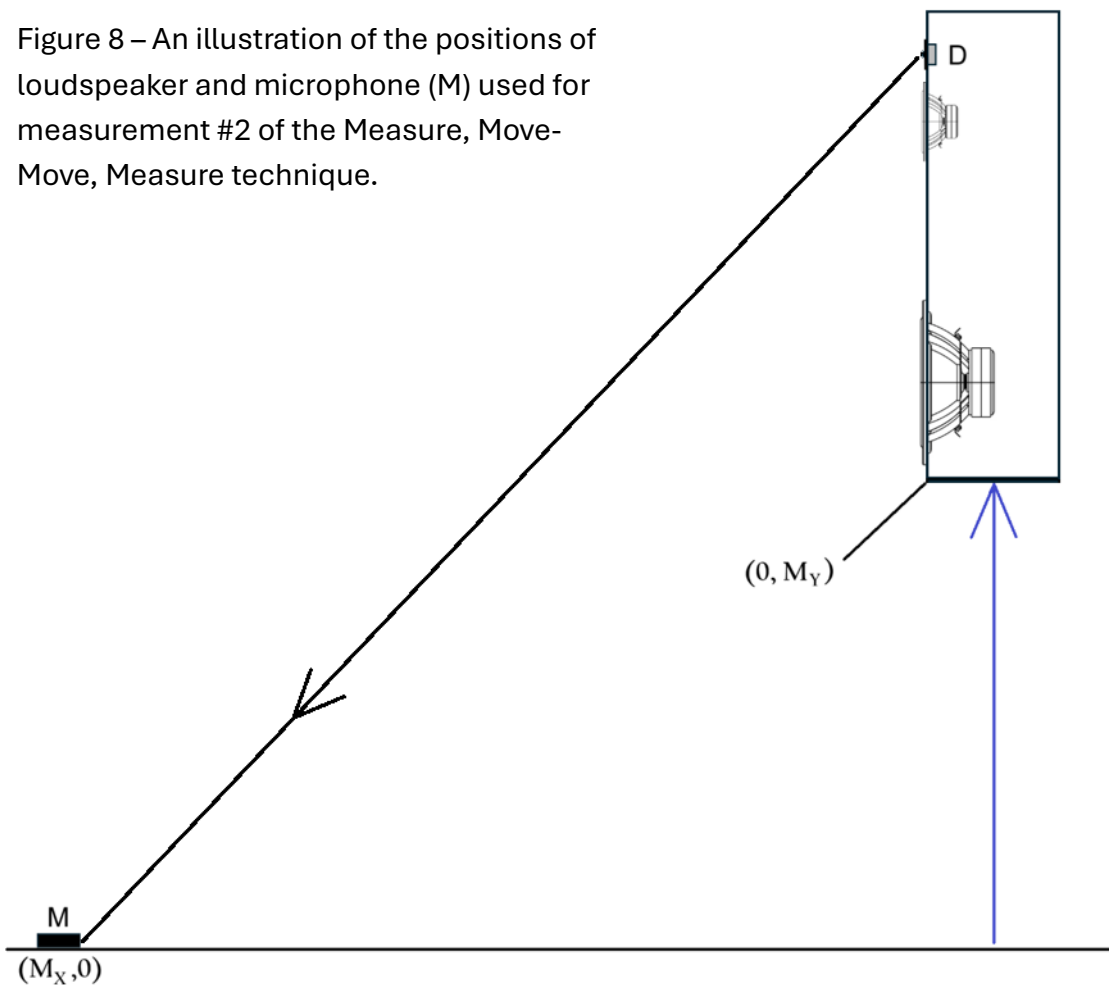


Figure 7 – A hypothetical position for the microphone (M), obtained after reflecting the microphone position from figure 6 about the groundplane (blue arrow).

position is now “underground”. Placing a microphone underground is not possible, however, some additional gymnastics will solve this problem. Keep the relative positions of microphone and loudspeaker the same and simultaneously move both upwards by the distance M_Y . This produces the physically realizable arrangement shown in figure 8. The microphone is now located on the groundplane at $(M_X, 0)$ and the loudspeaker is elevated above the ground by the distance M_Y . In practice one simply starts with the setup for measurement #1, moves the microphone straight down to the floor to position $(M_X, 0)$, and then places the loudspeaker on a stand so that it sits at position $(0, M_Y)$. Since we must move two objects, this inspired the “move-move” part of the name.

Figure 8 – An illustration of the positions of loudspeaker and microphone (M) used for measurement #2 of the Measure, Move-Move, Measure technique.



MEASUREMENT #2: The arrangement of microphone and loudspeaker is shown in figure 8. Similar to the setup for the “lift and tuck” measurement technique, we need to elevate the loudspeaker using a stand that is acoustically transparent but we do not “tuck” the loudspeaker towards the microphone but rather keep it in the same orientation that was used for measurement #1. The height that the loudspeaker is elevated for measurement 2 is **not arbitrary**. It must be **exactly equal** to M_Y , the height

of the microphone above the groundplane in measurement #1. Because measurement #2 is a groundplane measurement, the sound from the loudspeaker and its image in the groundplane do not interfere in the frequency domain except at relatively high audio frequencies, similar to the lift and tuck method.

In measurement #2, the microphone has been positioned at the exact same off-axis angle as the measurement #1 groundplane reflection and is at the same distance away. That's what we want but what is being measured exactly? Measurement #2 is a groundplane measurement. The microphone records the direct sound (as shown in figure 8) from the loudspeaker as well as its groundplane reflection (not shown). Since these signals are identical, they add coherently and the overall level is doubled.

Now we will calculate the free-field response. In the first step, the amplitude level of measurement #2 is reduced by half, e.g. by 6.02 dB (see note 1). The result is the output from the loudspeaker only, as set up for measurement #2. But since that is exactly the same as the off-axis angle and distance from the reflection produced by measurement #1 we can label it as such:

Reflection response, measurement #1 = $0.5 \times$ measurement #2

Recall from equation 1 that the total response is the direct response plus the reflection response:

$$p_{TOT} = p_D + p_R$$

Thus, to obtain the direct, or free-field response (p_D) we simply subtract the reflection response from the data collected in measurement #1 ($p_D = p_{TOT} - p_R$):

Free-field response = measurement #1 - $0.5 \times$ measurement #2.

The Measure, Move-Move, Measure technique has produced three different responses:

- In-room response including the groundplane reflection (p_{TOT})
- The groundplane reflection response (p_R)
- The direct or free-field response (p_D or p_{FF})

each of which provides slightly different information to the designer. The free-field response is useful for designing the crossover and for checking how well it integrates the drivers. The in-room response captures the baffle doubling effect and the lowest frequency interference dip while the groundplane reflection response provides clues about the loudspeaker's power response in the vertical plane. The latter two vary with the standoff distance but are useful when deciding how to voice the loudspeaker.

Because the measure, move-move, measure technique is a combination of direct and groundplane reflection measurements it will be less susceptible to high frequency measurement error than either the lift and tuck method or the Gander method. This is because the groundplane measurement (measurement #2) is made at a relatively high

off axis angle for the high frequency transducer of the loudspeaker (e.g. the tweeter). This is the same angle as for the reflected sound in measurement #1. At high off axis angles the tweeter will have less acoustic output due to source directivity leading to less high frequency energy in the reflection response. Most of the high frequency information in the free-field response will come from the direct sound (via measurement #1) and the amount of subtraction or correction from measurement #2 at high frequencies will be small. As a result, the error due to the heights of the source and microphone in measurement #2 (see the appendix in Gander) is minimized.

The Measure, Move-Move, Measure technique has advantages and disadvantages:

Advantages:

- Because it is primarily a direct rather than groundplane measurement it does not lose accuracy at high frequency like the Lift and Tuck or Gander methods.
- The loudspeaker can remain upright or in any orientation as long as that remains the same for measurements 1 and 2. Because the speaker can remain upright, off axis measurements in the horizontal plane are easily carried out by rotating the loudspeaker on a turntable.
- The technique produces not only the free-field loudspeaker frequency domain response, but also the response of the groundplane reflection as well as the sum of these, showing the interference pattern produced by the reflection. This extra information can be helpful in choosing how to voice the loudspeaker.
- There is no guesstimation about the baffle doubling effect. It is automatically captured during the measurement process.
- The technique works equally well at any standoff distance

Disadvantages:

- Requires two measurements instead of one
- Both measurement #1 and #2 require precise positioning of the mic and speaker

NOTE 1: The amplitude adjustments of 6.02 dB or by a factor of 0.5 only apply when the groundplane is perfectly specular (is a perfect acoustic reflector and $R=1$). When this condition is not met, use the procedure outlined in the next section to correct the measured data.

Accounting for an imperfect groundplane reflection

Both measurement techniques remain completely valid when the groundplane is not perfectly reflective. In that case, the groundplane reflection transfer coefficient R must be determined and then used to correct the measured responses as described below. Keep in mind that the pressure variables used in the formulas below are frequency-dependent complex numbers.

Determining the Groundplane Reflection Transfer Coefficient:

R is characterized using two measurements made on a small loudspeaker. For the first measurement, elevate it and the microphone a few meters above the ground and away from other boundaries. The microphone should be placed on the listening axis, at a distance of e.g. at least 1m to be in the far field. The data will include groundplane and/or other boundary reflections that should be gated out to produce a quasi-anechoic free-field or “direct sound” pressure response, p_D . Save the minimum phase response. For the second measurement, set up the same loudspeaker for the lift and tuck technique above the groundplane you wish to characterize. The conditions should be identical to the quasi-anechoic measurement (standoff distance, mic position, signal and drive voltage, use the same gating settings, etc.). Perform the measurement to obtain the pressure response p_{TOT} and again save the minimum phase response.

In this technique, the microphone simultaneously records both the direct sound and the reflection but, because the pathlengths are identical, the reflection signal is equal to the direct sound times R . Therefore, one can write:

$$p_{TOT} = p_D + p_R = p_D + R \times p_D = (1 + R) \times p_D$$

Solving for R we have:

$$R = p_{TOT} / p_D - 1 \tag{A1.1}$$

Except for the factor 1, the pressure values are complex numbers that are a function of frequency. Assuming that R is real valued, a shortcut method is simply to subtract the frequency-dependent SPL values to perform the division, e.g.

$$R \text{ in Decibels} = (\text{Total SPL}) - (\text{Direct SPL})$$

and then optionally converting from log (Decibel scale) to linear scale values:

$$R = 10^{(R \text{ in Decibels} / 20)}$$

Now that R has been determined, when measuring an unknown loudspeaker with one of the groundplane techniques you need to correct the measurement data as explained below.

Correcting Data Collected Using the Lift and Tuck Technique:

The pressure data p_{TOT} is recorded by the microphone, and we wish to know the free-field response, p_{FF} . Use the following transformation:

$$p_{FF} = p_{TOT} / (1 + R) \quad A1.2$$

You can see that if R is real and has a value of 1.0 (perfectly reflecting), the factor $1 + R$ simplifies to 2.0. It is important to remember that R may be frequency dependent.

If you wish to make the correction using the R value in Decibels, then

$$\text{Free Field SPL} = \text{Total SPL} - (R \text{ in Decibels}).$$

Correcting Data Collected Using the Measure, Move-Move, Measure Technique:

In this technique the end goal of measurement #2 is to produce data representing the reflection component recorded in measurement #1. Although measurement #2 is essentially the Lift and Tuck technique but without the tuck, we want to do a slightly different type of correction. This is illustrated by the following two steps. First, if we perform the same correction used for the Lift and Tuck method:

$$p_{FF} = p_{TOT(2)} / (1 + R)$$

where $p_{TOT(2)}$ is the response obtained via measurement #2, the correction completely removes the effect of the non-ideal groundplane reflection for this measurement geometry. p_{FF} is the free-field pressure at the off-axis angle used in measurement #2 but that is not exactly the response that we are trying to generate. By multiplying by R we add the effect of the groundplane reflection back in. The resulting value is the reflection component of measurement #1, $p_{R(1)}$:

$$p_{R(1)} = p_{TOT(2)} \times R / (1 + R) \quad A1.3$$

You can see that if R is real and equal to 1.0 (perfectly reflecting), the factor $R / (1 + R)$ simplifies to 0.5.

After calculating $p_{R(1)}$ we can proceed to subtract it from the data collected in measurement #1 to obtain the free field response:

$$p_{FF} = p_{TOT(1)} - p_{R(1)} \quad A1.4$$

where $p_{TOT(1)}$ is the response obtained via measurement #1.

The datasets p_{FF} , $p_{TOT(1)}$, and $p_{R(1)}$ each contain different information about the loudspeaker system response that can be useful for the designer/developer.

How to Make Successful Measurements on Irregular or Soft Surfaces

For good low frequency resolution, a large area away from obstacles is best, and outdoor locations such as a vacant parking lot, street, or even a large, maintained grass lawn can be ideal under certain conditions. At the same time, for good high frequency accuracy the surface should be quite smooth and homogeneous, lacking voids, etc. Irregular surface roughness and porosity will cause diffraction and/or scattering of the incident wave, which will produce errors at middle and high frequencies. For example, newly laid aggregate and asphalt paving can have small surface cavities due to the irregular nature of the aggregate when it is not completely filled in by the asphalt component. Pre-cast cement paver stones often have a porous fraction and are of a lower density than poured concrete. A grass lawn is not homogeneous on small length scales, and the underlying ground surface may be irregular. All of these materials can be improved for acoustic measurements by adding a plate or layer of another material above them.

The overlaying layer should have the following characteristics:

- 1) The material should be relatively thin. Thick material will protrude above the ground level and may produce diffraction along the edges.
- 2) Keep the material in intimate contact with the ground below.
- 3) Choose a material with similar acoustic impedance compared to the ground below. On top of asphalt, a thin, smooth metal plate will work well, however, if the surface was grass and a hard metal or glass plate is used, diffraction will be generated around the periphery because of the large change in acoustic impedance between the two materials. For covering grass, cut the grass as short as possible, fill and level uneven areas with soil, and then cover the surface with 2-3 mm thick needle felted polyester panels or fabric. This has an even surface and an acoustic impedance similar to the ground below. The material must allow for airflow – do not use an acoustically opaque material. Press down evenly to create intimate contact.
- 4) The larger the coverage, the better. At a minimum, cover the area over which the groundplane image will be observed from the perspective of the microphone to ensure that all parts of the image are reflected in the same way.
- 5) Where the added layer of material ends, the ground should be as flat as possible. For example, on grass you should cut the grass very short in the area immediately surrounding the plate. If making measurements on a street, keep as far away as possible from raised curbs. Any solid object or structure nearby will contaminate the measurement with a reflection that will limit the measurement accuracy.

Anytime you add an overlayer to a soft ground, the groundplane transfer function must be characterized to ensure that subsequent measurements are accurate.

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Appendices:

Tips for the Measure, Move-Move, Measure Technique

Tip #1: A stand is needed to elevate the loudspeaker for measurement #2 by exactly the height that the microphone was above the ground for measurement #1. There is an easy way to build an acoustically transparent stand that can be used during both measurements, first for the microphone and then for the loudspeaker. Buy flat gabion cage panels or heavy duty welded wire mesh. Build a circa 1m or 4 foot tall stand from these materials. For measurement #1, the loudspeaker should be placed on the floor and the microphone on the top of the stand. Elevate the loudspeaker using a supplemental stand or box so that the tweeter or listening axis is brought up to the 1m microphone elevation. For measurement #2, take the mic off of the stand and put it on the floor. Place the supplemental stand (if used) on top of the wire stand and then put the loudspeaker on top of that. Measure again to collect your measurement #2 data.

Tip #2: It is very important that, between measurements 1 and 2, there is only vertical movement of the microphone and loudspeaker. To accomplish this, before making any measurements I mark out two points on the floor – one is the (0, 0) “origin” where the DUT will be located and other is the reference point for the microphone at the desired standoff distance, e.g. (M_x , 0). When I want to position something directly over one of these points, I use a laser level that has up/down laser “dots” that function as plumb lines. I place the level directly over the mark I made on the floor and position the mic or loudspeaker so that the “dot” that is being projected straight up illuminates a reference spot on the object, e.g. the tip of the microphone, or the center of the bottom edge of the loudspeaker baffle. During measurement #1, the mic will be positioned at some height (e.g. 1m) above its reference mark, and the loudspeaker sits on the floor with its baffle aligned with the origin reference mark. For measurement #2 the mic pickup is in contact with the floor on its mark, and the bottom front edge, at center, of the loudspeaker baffle is illuminated with the laser dot, directly above the floor mark while on top of the 1m tall stand. The standoff distance M_x between the two floor marks is not important but should remain the same for measurements 1 and 2.

Data Processing: The SPL adjustment and subtraction can be performed equally well in the time domain or the frequency domain. In the time domain, the required operations can be done rapidly in ARTA. The amplitude (or gain) of the impulse response from measurement #2 is multiplied by the factor 0.5 (see note 1) and the polarity reversed. The result is then summed with the impulse response from measurement #1, gated, and used to calculate the free-field frequency and phase responses. To do the manipulations in the frequency domain, identical gating is applied to measurements 1 and 2 and both are converted to the frequency domain. The SPL of measurement #2 is reduced by 6.02 dB (see note 1), its polarity reversed, and then the two complex frequency domain responses are added together. Keep in mind that the time domain processing is much more sensitive to positioning errors between measurements 1 and 2 compared to the frequency domain processing method.

A computer program for correcting groundplane measurements

The following Python code performs the calculations from Appendix 1 that correct measurement data for non-ideal groundplane reflectivity. Input and output data files must be in the FRD file format. Input and output filenames as well as the calculation type (1,2,3,4) must be supplied by the user using command line flags and parameters. The `--help` option prints syntax and usage information. The calculations performed are:

Type 1 : Calculate the Groundplane Transfer Function as $\text{infile1} / \text{infile2} - 1$ as shown in equation A1.1

Type 2 : Calculate the Free-Field Response from the Lift and Tuck Data in `infile1` using the Groundplane Transfer Function in `infile2` as shown in equation A1.2

Type 3 : Calculate the Reflection Response from the MMMM Measurement 2 Data in `infile1` using the Groundplane Transfer Function in `infile2` as shown in equation A1.3

Type 4 : Calculate the Free-Field Response from the MMMM Measurement 1 Data in `infile1` by subtracting the Reflection Response in `infile2` as shown in equation A1.4

Before Use, make sure that Python and the required modules are installed:

```
sudo apt update; sudo apt -y install python3 python3-pandas python3-numpy
```

```
import argparse, textwrap
import numpy as np
import pandas as pd

def load_frd(filepath):
    """Reads an FRD file and returns frequency and complex pressure."""
    # Read file ignoring lines starting with * (comments)
    df = pd.read_csv(
        filepath,
        engine='python',
        sep=r"\s+",
        comment="*",
        header=None,
        names=["Freq", "Mag_dB", "Phase_deg"],
    )

    # Convert dB to linear magnitude
    mag_lin = 10 ** (df["Mag_dB"] / 20)
    # Convert phase from degrees to radians
    phase_rad = np.radians(df["Phase_deg"])

    # Create complex number: text = mag * e^(j * phase)
    complex_val = mag_lin * np.exp(1j * phase_rad)

    return df["Freq"].values, complex_val

def save_frd(filepath, freq, complex_val):
    """Converts complex values back to Mag(dB)/Phase(deg) and saves as FRD."""
    mag_lin = np.abs(complex_val)
    # Avoid log10(0) by replacing zeros with a tiny number
    mag_lin = np.where(mag_lin == 0, 1e-12, mag_lin)

    mag_dB = 20 * np.log10(mag_lin)
    phase_deg = np.degrees(np.angle(complex_val))

    # Write to file with standard FRD headers
    with open(filepath, "w") as f:
        f.write("* Frequency(Hz) Magnitude(dB) Phase(deg)\n")
        for freq, mag, phase in zip(freq, mag_dB, phase_deg):
            f.write(f"{freq:<12.4f} {mag:<12.4f} {phase:<12.4f}\n")
```

```

def align_and_interpolate(freq_target, freq_source, p_source):
    """
    Checks if frequencies align. If not, interpolates source data (Mag/Phase)
    onto the target frequency grid using numpy.interp.
    """
    if np.allclose(freq_target, freq_source):
        return p_source
    print("Interpolating FILE2 data to the FILE1 frequency grid...")

    # Deconstruct complex values back to dB and Phase for stable interpolation
    mag_dB = 20 * np.log10(np.abs(p_source))
    phase_deg = np.degrees(np.angle(p_source))
    # Unwrap phase component, phase in degrees
    phase_deg = -np.unwrap(-phase_deg, period=360)
    # Perform linear interpolation directly using np.interp
    mag_dB_interp = np.interp(freq_target, freq_source, mag_dB)
    phase_deg_interp = np.interp(freq_target, freq_source, phase_deg)
    # Wrap the interpolated phase back to the [-180, 180) interval
    phase_deg_interp = ((phase_deg_interp + 180) % 360) - 180
    # Reconstruct complex numbers on the target frequency grid
    mag_lin_interp = 10 ** (mag_dB_interp / 20)
    phase_rad_interp = np.radians(phase_deg_interp)
    print("    ...interpolation complete.")
    #return the complex pressure variable
    return mag_lin_interp * np.exp(1j * phase_rad_interp)

def main():
    parser = argparse.ArgumentParser(
        description="This program performs various corrections to data generated using groundplane-based
        acoustic measurement techniques.",
        usage='use "python %(prog)s --help" for more information',
        formatter_class=argparse.RawTextHelpFormatter
    )
    # File arguments
    parser.add_argument(
        "--infile1", "--f1", type=str, required=True, help="Path and Filename of the first input file."
    )
    parser.add_argument(
        "--infile2", "--f2", type=str, required=True, help="Path and Filename of the second input file."
    )
    parser.add_argument(
        "--outfile", "--of", type=str, required=True, help="Path and Filename of the output file."
    )
    # Calculation type argument with restricted choices
    parser.add_argument(
        "--calc_type", "-c",
        type=int,
        required=True,
        choices=[1, 2, 3, 4],
        help=textwrap.dedent('''\
        Indicate the type of calculation to perform by number:
        1 : Calculate the Groundplane Transfer Function as infile1 / infile2 - 1
        2 : Calculate the Free-Field Response from the Lift and Tuck Data in infile 1 using the
        Groundplane Transfer Function in infile2
        3 : Calculate the Reflection Response from the MMMM Measurement 2 Data in infile1 using the
        Groundplane Transfer Function in infile2
        4 : Calculate the Free-Field Response from the MMMM Measurement 1 Data in infile1 by subtracting
        the Reflection Response in infile2
        NOTE: All input and output files should be in the FRD file format.
        ''')
    )
    args = parser.parse_args()

```

```

try:
    # Load the files and return frequency
    # Returns the pressure as a complex value
    freq_file1, file1_pressure = load_frd(args.infile1)
    freq_file2, file2_pressure = load_frd(args.infile2)

    # Align file2_pressure to the freq_file1 frequency grid (if necessary)
    # Returns a complex variable
    file2_pressure = align_and_interpolate(freq_file1, freq_file2, file2_pressure)

    # Route execution based on the chosen calculation type
    if args.calc_type == 1:
        print(f"Calculating the Groundplane Transfer Function from files {file1} and {file2}")
        print(f"Performing the calculation: RESULT = FILE1 / FILE2 - 1")
        result_complex = file1_pressure / file2_pressure - 1.0
        message = f"The Groundplane Transfer function will now be saved to the file {args.file2}."
    elif args.calc_type == 2:
        print(f"Calculating the Free-Field Response from the Lift and Tuck Data in {args.infile1}
using the Groundplane Transfer Function in {args.infile2}")
        print(f"Performing calculation: RESULT = FILE1 / (1 + FILE2)")
        result_complex = file1_pressure / (1.0 + file2_pressure)
        message = f"The Free-Field Data will now be saved to the file {args.outfile}."
    elif args.calc_type == 3:
        print(f"Calculating Reflection Response from the MMMM Measurement 2 Data in {args.infile1}
using the Groundplane Transfer Function in {args.infile2}")
        print(f"Performing the calculation: RESULT = (FILE1 * FILE2) / (1 + FILE2)")
        result_complex = (file1_pressure * file2_pressure) / (1.0 + file2_pressure)
        message = f"The Reflection Data will now be saved to the file {args.outfile}."
    elif args.calc_type == 4:
        print(f"Calculating the Free-Field Response from the MMMM Measurement 1 Data in
{args.infile1} by subtracting the Reflection Data in {args.infile2}")
        print(f"Performing the calculation: RESULT = FILE1 - FILE2")
        result_complex = file1_pressure - file2_pressure
        print(f"The Free-Field Data will now be saved to the file {args.outfile}.")

    # Save result to file
    save_frd(args.outfile, freq_file1, result_complex)

except Exception as e:
    print(f"Error: {e}")

if __name__ == "__main__":
    main()

```